

I_G → Gini Impurity

very similar to entropy

sklearn
↑

Entropy or Gini

$Y \rightarrow Y_k$

$$I_G(Y) = 1 - \sum_{i=1}^K (P(Y=i))^2$$

$C = 0.1, 0.001$

↓

eg. Table ten

↓

$$I_G(Y) = 1 - \left[\left(\frac{\text{yes}}{2} \right)^2 + \left(\frac{\text{no}}{2} \right)^2 \right]$$

$$I_G(Y) = x_1$$

Case i

$$p(Y+) \rightarrow 0.5$$

$$p(Y-) \rightarrow 0.5$$

log

()³

(Case ii)

$$p(Y+) = 1$$

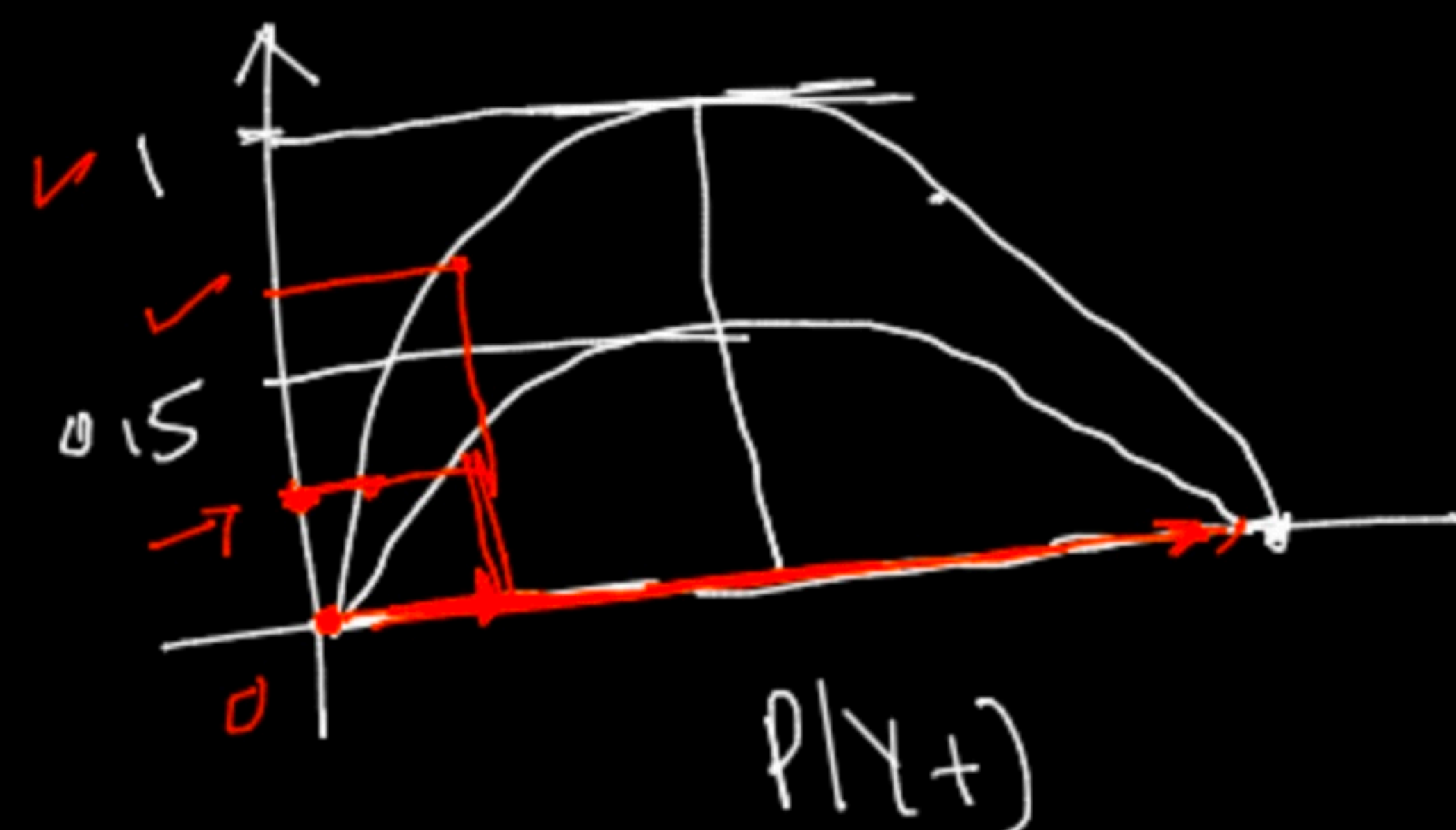
$$p(Y-) = 0$$

$$I_4(Y) = 1 - (1 - 0) = 0$$

$$I_4(Y) = 1 - [0.125 + 0.25]$$

$$= 1 - 0.5$$

$$I_4(Y) = 0.5$$



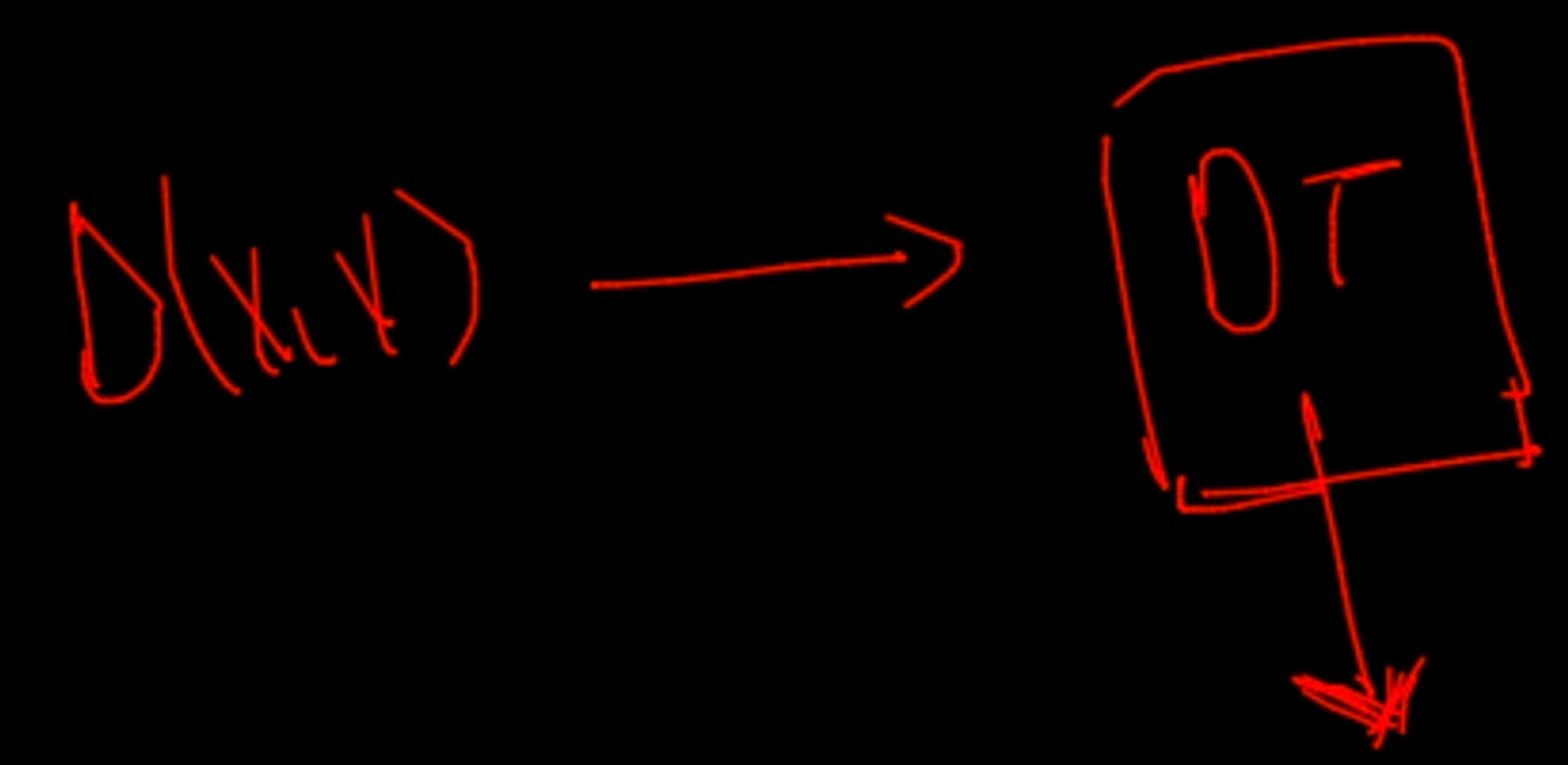
$$p(Y+) = 0.2, p(Y-) = 0.8$$

Computing 'log' took more time complexity as compare compare

small

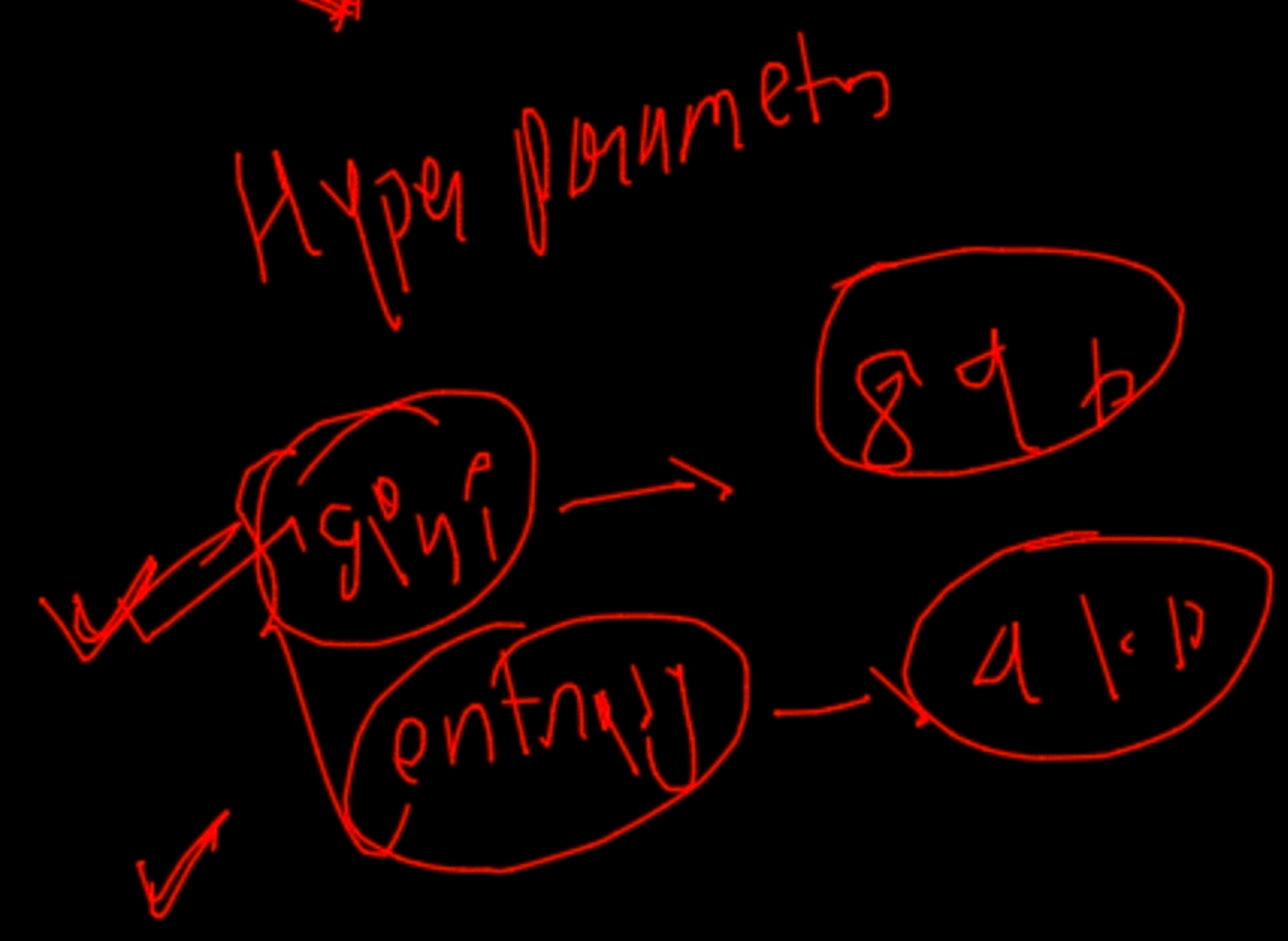
Huge

factor



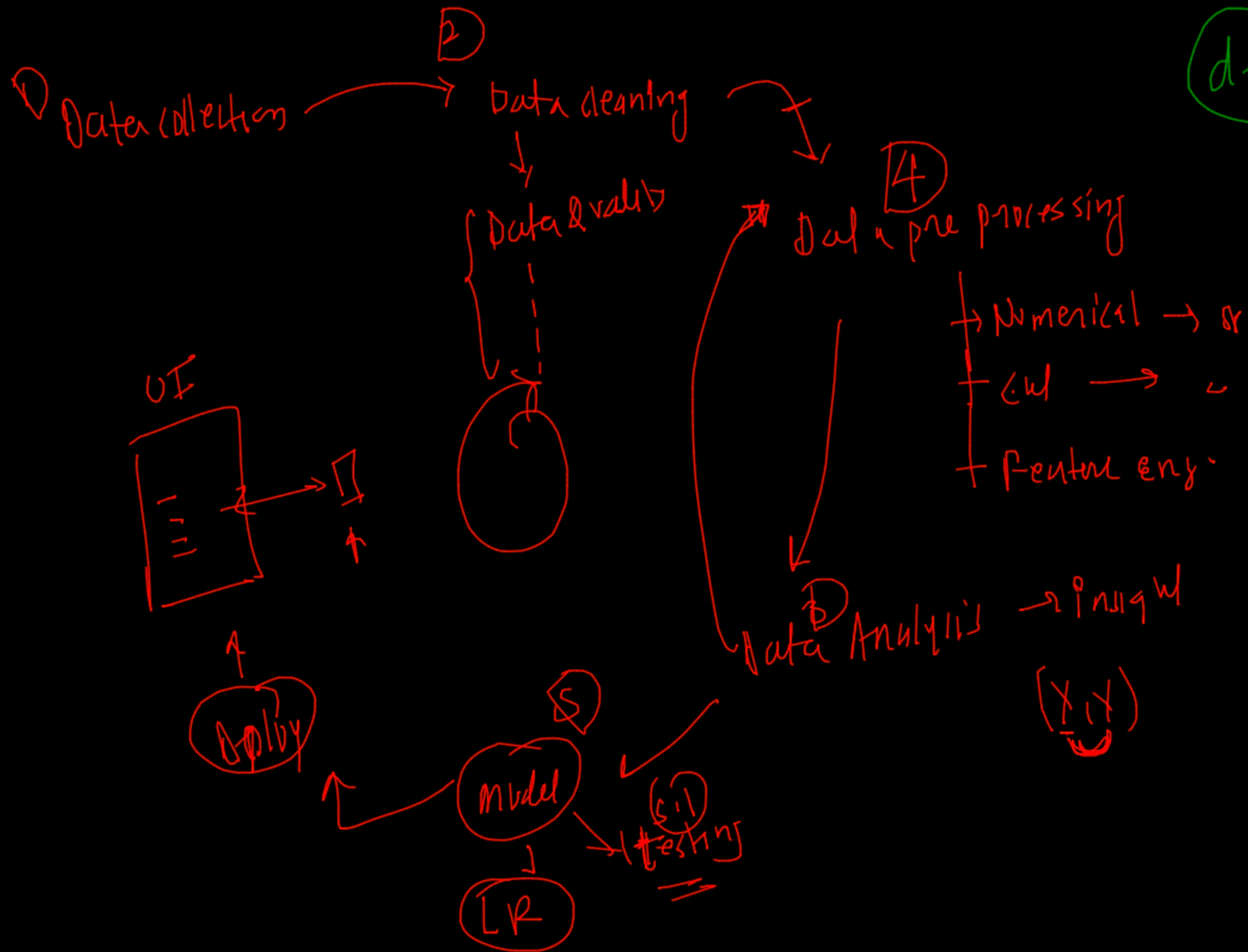
① Data size

↓
under small
(2-lakh)



15 lakh

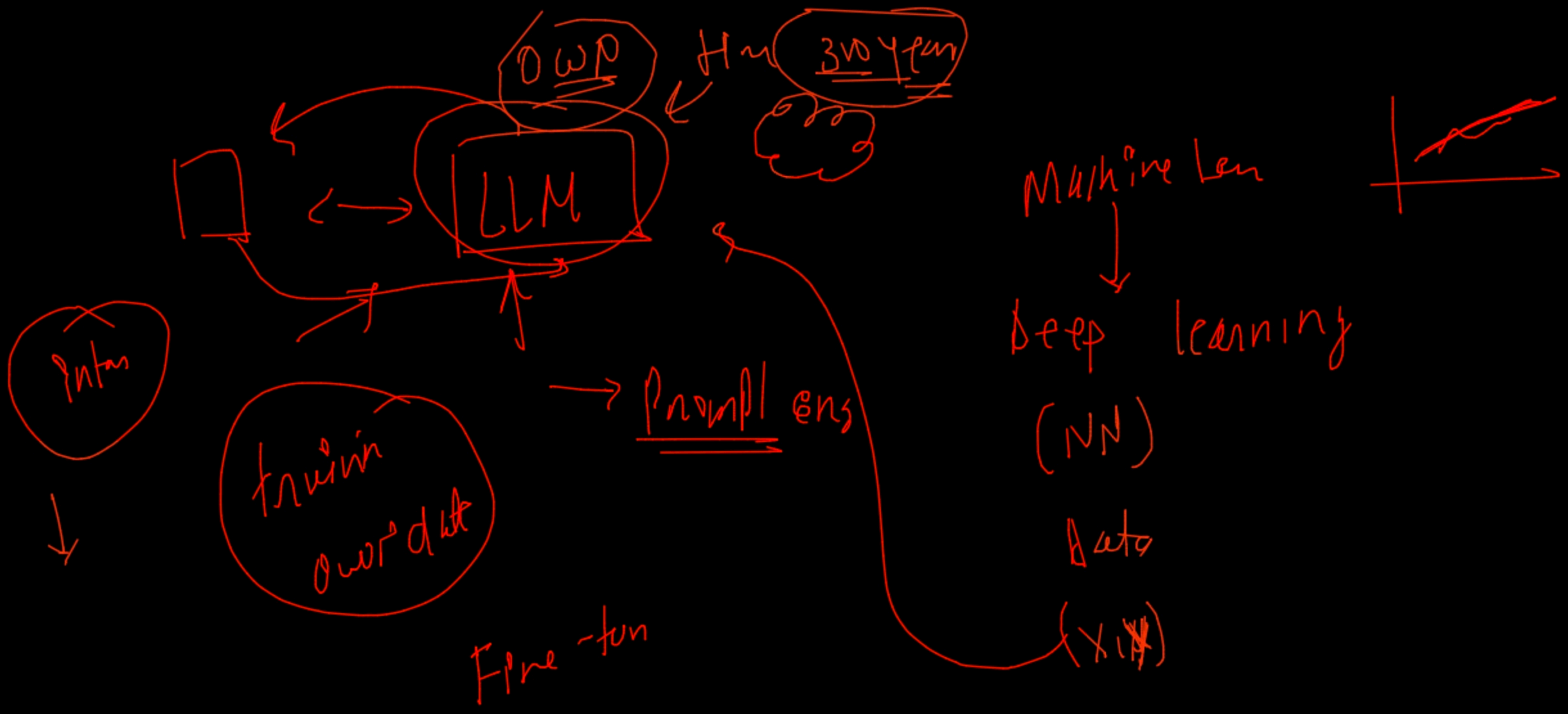




depth ↑
↓ overfitting

min sample
↓
5/5/10

=====
=====
=====
=====



WLP

← weltm

python

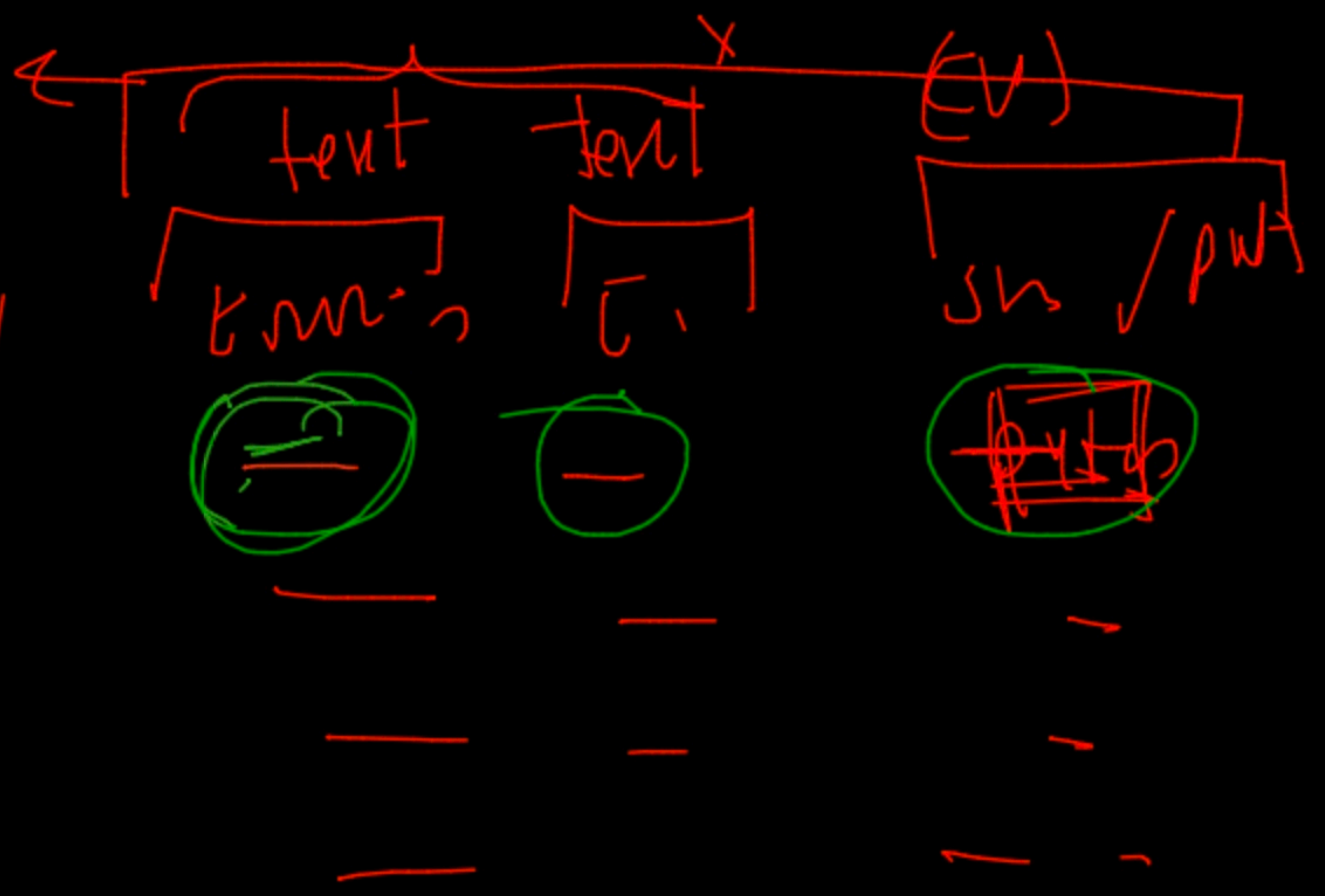
HTML

CSV

(QML)

Tic

123



API 4D

B64
↑
(HTML)

210

BeautifulSoup

text + image

Multi model

library

10

LXM

gpt

ML

DL

RNN LSTM

text

NLP

vision

CV

100%